

Bangladesh National Election 2026: Big Data Correlation and Predictive Analytics Using Apache Spark

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Abstract—This paper presents a comprehensive big data analytics pipeline for the 13th Bangladesh National Parliamentary Election held on February 12, 2026, conducted concurrently with the July Charter constitutional referendum. Leveraging Apache Spark, Spark SQL, and Spark MLlib on Google Colab, we integrate constituency level election results with granular socio-economic indicators derived from the Bangladesh Bureau of Statistics (BBS) 2022 Population and Housing Census. Our data engineering pipeline processes over 95,000 raw demographic records across four census dimensions and condenses them into 269 ML-ready constituency profiles via a custom geospatial mapping bridge and K-Nearest Neighbors (KNN) imputation. The analytical pipeline encompasses: (i) Pearson and Spearman dual correlation analysis revealing strong negative associations between internet penetration and voter turnout ($r = -0.5948$); (ii) regression modeling achieving $R^2 = 0.6302$ for voter turnout prediction; (iii) Random Forest classification of winning coalitions attaining 76.79% accuracy with AUC-ROC of 0.7738; and (iv) K-Means clustering identifying two dominant constituency archetypes (Silhouette = 0.7078). Extended hypothesis testing across 13 socio-political hypotheses reveals a U-shaped wealth radicalism relationship, statistically significant coalition-referendum cleavages ($p = 0.000781$), and a border radicalization effect (OR = 1.90). Youth demographics emerge as the primary structural determinant of electoral outcomes.

Index Terms—Big Data Analytics, Apache Spark, MLlib, Bangladesh Election 2026, Random Forest, K-Means Clustering, Electoral Prediction, Socio-Economic Analysis

I. INTRODUCTION

The 13th Bangladesh National Parliamentary Election, held on February 12, 2026, featured a concurrent constitutional referendum on the “July Charter.” The election was dominated by two coalition blocs: the **BNP Coalition** (212 seats) and the **Jamaat/Reform Coalition** (77 seats), with 8 seats captured by independents.

Despite the availability of granular census data, no systematic big data analysis has quantified the relationship between socio-economic indicators and electoral outcomes. This project addresses four research questions: (RQ1) What are the statistically significant correlations between socio-economic indicators and electoral outcomes? (RQ2) Can regression models predict voter turnout from census data? (RQ3) Can

a classifier predict the winning coalition from demographics alone? (RQ4) Do constituencies cluster into distinct socio-economic archetypes with divergent voting patterns?

We implement a complete end-to-end pipeline using Apache Spark (PySpark) on Google Colab, spanning data ingestion, geospatial mapping, feature engineering, imputation, correlation analysis, regression, classification, clustering, and extended hypothesis testing.

II. DATA SOURCES AND ENGINEERING

A. Data Sources

The project integrates two primary data families:

BBS Census 2022: Four thematic sheets were extracted—SDG Tracking (Internet, Clean Fuel, Sanitation, Financial Inclusion), Age-Structure Demographics (Youth/Senior Electorate, First-Time Voters, NEET Youth), Gender Demographics (Female Population), and Housing Structure (Pucca/Brick housing, Extreme Poverty proxy).

Election Commission Results: Constituency-level results for all 300 seats, including candidate vote counts, party mappings, and July Charter referendum YES/NO percentages (available for 168 constituencies).

Constituency Boundaries: A PDF listing 300 parliamentary boundaries was parsed using `pdfplumber` for the geospatial bridge.

B. Data Engineering Pipeline

The pipeline transforms over 95,000 raw census records into 269 ML-ready constituency profiles through seven stages:

Stages 1–2: Ingestion and Feature Extraction. BBS Excel files were loaded via `pd.read_excel()` with schema sanitization. Four feature families were extracted: SDG indicators, youth/age demographics (with Voting Age Population correction excluding ages 0–14), gender ratios, and housing/wealth metrics. The Wealth Polarization Index was computed as $WPI_Log = \log(\text{Extreme_Poverty}/\text{Pucca})$.

Stage 3: Grand Merge. Feature families were fused using a LEFT JOIN to preserve 44 constituencies with missing SDG data.

TABLE I: Engineered Feature Set

Feature	Source / Description
Youth_Electorate_Pct	BBS C-05: Electorate aged 18–35
Senior_Electorate_Pct	BBS C-05: Electorate aged 60+
First_Time_Voter_Pct	BBS C-05: Electorate aged 18–22
NEET_Youth_Pct	BBS C-05: Youth not in edu/work
Female_Pop_Pct	BBS Gender: Female share
Pucca_Pct	BBS C-14: Brick housing (wealth)
Extreme_Poverty_Pct	BBS C-14: Jhupri+Kancha housing
WPI_Log	Engineered: $\log(\text{Poverty}/\text{Pucca})$
Internet_Pct	BBS SDG: Internet access
Clean_Fuel_Pct	BBS SDG: Clean cooking fuel
Sanitation_Pct	BBS SDG: Sanitation access
Organized_Learning_Pct	BBS SDG: Education participation
Financial_Inclusion_Pct	BBS SDG: Financial services
Urban_Proxy	Engineered: Urbanization index
Digital_Youth_Index	Engineered: $\text{Internet} \times \text{Youth}$
Infrastructure_Comp.	Engineered: Sanitation+Fuel
Deprivation_Index	Engineered: Poverty+NEET
Cluster_ID	K-Means: Constituency archetype

Stages 4–5: Geospatial Bridge. The BBS administrative hierarchy (Districts/Upazilas/Thanas) was mapped to EC constituency numbers via: (i) NLP fuzzy matching of location names against boundary descriptions, (ii) hardcoded city thana-to-constituency maps for Dhaka, Chattogram, Khulna, and (iii) district alias dictionaries for spelling discrepancies. Coverage achieved: **269 of 300 constituencies (89.6%)**.

Stage 6: KNN Imputation. Twelve districts (75 seats) lacked SDG sheets. `KNNImputer` ($k = 5$, distance-weighted) estimated missing values from the 5 most demographically similar constituencies, preserving multivariate correlation structure.

Stage 7: Election Results Cleaning. Vote count strings were parsed, a known anomaly corrected, party names translated via reference CSV, and fringe candidates removed.

C. Feature Engineering

Table I summarizes the 18 features used in the ML pipeline.

III. SYSTEM ARCHITECTURE

The pipeline was executed on Google Colab using PySpark for distributed processing, Spark SQL for structured queries, and Spark MLlib for all machine learning tasks (`Correlation`, `LinearRegression`, `RandomForestClassifier`, `RandomForestRegressor`, `KMeans`, `StandardScaler`, `VectorAssembler`, `StringIndexer`). `scikit-learn` was used exclusively for `KNNImputer`. Visualizations were generated with Matplotlib and Seaborn via Pandas conversion.

IV. EXPLORATORY DATA ANALYSIS

EDA was performed on the enriched dataset (`grand_df_enriched`) to understand distributional properties. Fig. 1 presents the target variable distributions, feature distributions, and full correlation matrix.

TABLE II: Key Correlation Results

Variables	Pearson	Spearman	Result
Internet vs. Turnout	−0.5948	−0.5392	Confirmed
Wealth Pol. vs. Margin	+0.0405	+0.0109	Rejected
NEET vs. Turnout	+0.2460	+0.0720	Rejected
Poverty vs. Turnout	+0.3228	+0.1857	Moderate
Urban vs. Turnout	−0.5111	−0.4482	Strong

TABLE III: K-Means Cluster Profiles ($K = 2$, Silhouette = 0.7078)

Feature	Cluster 0 ($N = 235$)	Cluster 1 ($N = 34$)
Pucca_Pct	16.41%	63.53%
Extreme_Poverty_Pct	66.18%	15.82%
Internet_Pct	26.04%	49.46%
Clean_Fuel_Pct	10.79%	77.75%
Voter_Turnout_Pct	61.79%	49.11%

V. CORRELATION ANALYSIS

Pearson and Spearman correlation matrices were computed using `pyspark.ml.stat.Correlation` on $N = 268$ constituencies. Table II presents hypothesis-driven results.

The strongest finding was a moderate-to-strong negative relationship between internet penetration and voter turnout ($r = -0.5948$). Among 168 constituencies with referendum data, all socio-economic correlations with YES vote percentage were weak ($|r| < 0.11$), motivating non-linear classification models.

VI. K-MEANS CLUSTERING ANALYSIS

K-Means clustering was applied to discover latent socio-economic archetypes, independent of electoral outcomes. All 12 features were standardized via `StandardScaler`. The Silhouette Score was computed for $K \in \{2, \dots, 10\}$ using `ClusteringEvaluator`.

$K = 2$ achieved the highest Silhouette Score of **0.7078**, indicating exceptionally strong cluster separation. Table III profiles the two archetypes.

Cluster 0 (“Rural/Deprived,” 87.4%) exhibits high poverty, low connectivity, and *higher* turnout (61.79%). Cluster 1 (“Urban/Affluent,” 12.6%) captures metropolitan constituencies with high wealth and *lower* turnout (49.11%)—a 12.68 percentage point gap.

VII. REGRESSION ANALYSIS

Two regression models were trained (Train: $N = 216$, Test: $N = 52$):

Model A (Voter Turnout): Linear Regression with ElasticNet regularization, combining L1 feature selection and L2 multicollinearity handling.

Model B (Vote Share): Random Forest Regressor for non-linear relationships.

Model A achieves $R^2 = 0.6302$, explaining 63% of turnout variance. The top predictors are `Youth_Electorate_Pct` (coeff = -5.797) and `Senior_Electorate_Pct` (coeff = -5.777), both negative—higher demographic concentration of either age

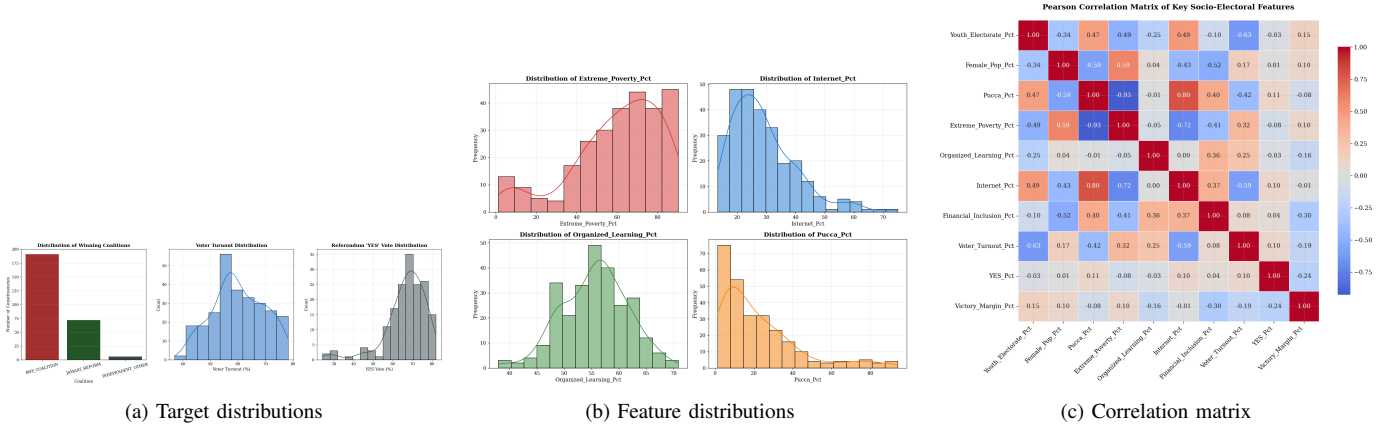


Fig. 1: Exploratory Data Analysis: (a) Coalition victory and turnout distributions, (b) univariate socio-economic feature distributions, (c) Pearson correlation matrix across 269 constituencies.

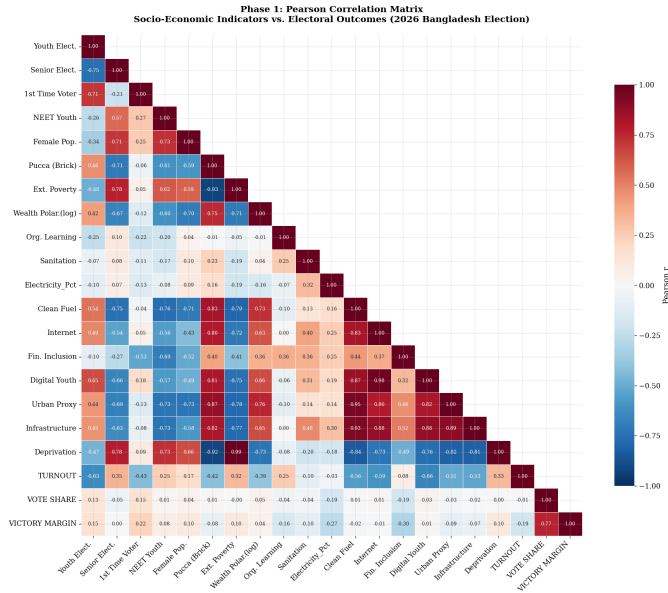


Fig. 2: Pearson correlation heatmap with electoral outcome variables positioned at bottom-right ($N = 268$).

TABLE IV: Regression Performance ($N_{\text{test}} = 52$)

	Target	R^2	RMSE	MAE
A	Turnout	0.6302	5.23%	4.44%
B	Vote Share	-0.0051	9.54%	6.94%

extreme *decreases* turnout. Model B achieves $R^2 = -0.0051$, confirming that vote share is unpredictable from census data—it reflects campaign dynamics rather than structural conditions.

VIII. CLASSIFICATION MODEL

A. Setup

The capstone analysis predicts coalition victory (BNP vs. Jamaat/Reform) from census data. The minor INDEPENDENT_OTHER class (6 seats) was excluded. Inverse-

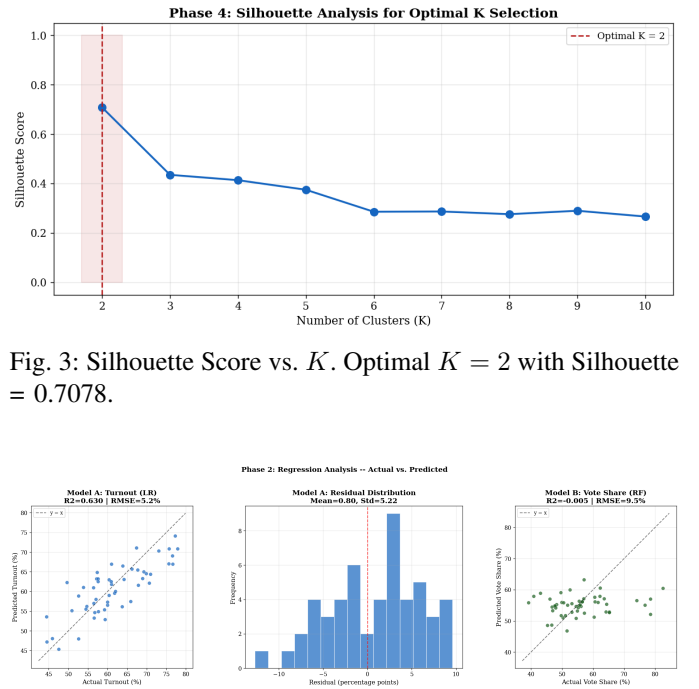


Fig. 4: Actual vs. Predicted plots: Model A (Turnout, $R^2 = 0.63$) and Model B (Vote Share, $R^2 \approx 0$).

frequency class weighting was applied: $w_{\text{JAMAAT}} = 1.8264$, $w_{\text{BNP}} = 0.6885$. The K-Means Cluster_ID was included as a feature. A stratified 80/20 split yielded Train: $N = 207$, Test: $N = 56$. A Random Forest Classifier with 200 trees was trained.

B. Results

The model surpasses the majority-class baseline by 4.16 percentage points. Fig. 6 shows the confusion matrix and feature importance ranking.

TABLE V: Classification Performance ($N_{\text{test}} = 56$)

Metric	Value	Class	P	R	F1
Accuracy	0.7679	BNP	0.81	0.87	0.84
W-F1	0.7609	Jamaat	0.64	0.53	0.58
AUC-ROC	0.7738				
Baseline	0.7262				

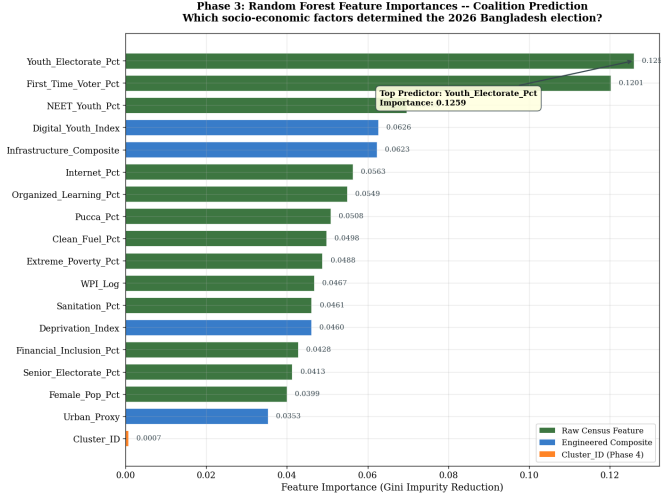


Fig. 5: Feature importance ranking. Green: raw BBS features; Blue: engineered composites; Amber: K-Means Cluster_ID.

C. Feature Importance

The top features are Youth_Electorate_Pct (0.1259) and First_Time_Voter_Pct (0.1201), together accounting for 24.61% of discriminative power. The top 5 features cumulatively explain 44.05% of Gini decrease. This establishes **youth demographics as the primary structural predictor of coalition victory**.

D. Ablation Study

Removing Cluster_ID reduced accuracy from 0.7679 to 0.7500 ($\Delta = +0.0179$) and F1 from 0.7609 to 0.7453 ($\Delta = +0.0156$), confirming the unsupervised-to-supervised pipeline linkage provides genuine predictive value.

IX. ADVANCED ANALYTICS PIPELINE

An extended pipeline was developed on the $N = 168$ constituency subset with referendum data.

A. Extended Regression

Referendum support prediction achieved $R^2 = 0.2836$ (RMSE = 7.9103), indicating that socio-economic features explain approximately 28% of YES vote variance. Winning margin prediction yielded $R^2 = 0.0162$ (RMSE = 15.8063), confirming margin unpredictability.

B. Extended Classification

The referendum outcome classifier achieved **90.00% accuracy** with AUC-ROC = **0.9136**, demonstrating that referendum

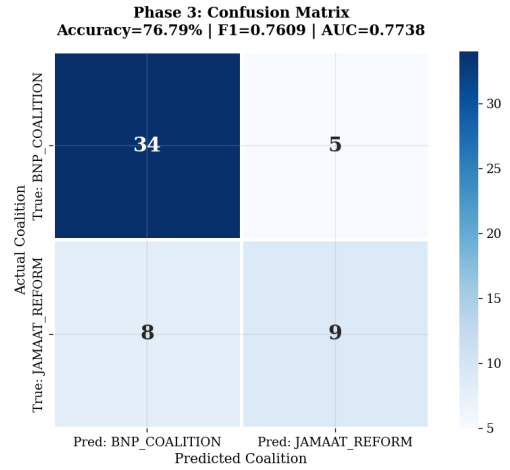
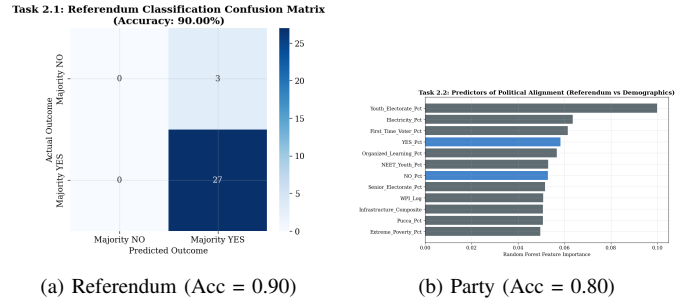


Fig. 6: Confusion matrix for Random Forest coalition classifier.



(a) Referendum (Acc = 0.90)

(b) Party (Acc = 0.80)

Fig. 7: Extended classification confusion matrices.

outcomes are highly predictable via non-linear ensemble methods. Multi-class party prediction achieved 80.00% accuracy (F1 = 0.7643), and voter behavior archetype classification achieved 80.00% across 4 categories.

C. Political Alignment Clustering

K-Means with $K = 4$ on political outcome features (Silhouette = 0.3776) identified four political tribes: anti-reform strongholds (NO = 62.94%, high margin), pro-reform dominance seats (YES = 68.16%), competitive high-turnout constituencies (margin = 12.18%, turnout = 66.49%), and competitive low-turnout urban seats.

X. DEDUCTIVE AND ADVANCED HYPOTHESIS TESTING

Thirteen hypotheses were tested using quintile analysis, t-tests, correlation, feature importance, and logistic regression. Table VI summarizes the ten deductive hypotheses.

A. H1: The Elite/Wealth Reversal

Jamaat win rates by Pucca_Pct quintile: Q1 = 37.74%, Q2 = 21.15%, Q3 = 15.09%, Q4 = 25.00%, Q5 = 37.74%. The identical rates at Q1 and Q5 confirm a U-shaped “horseshoe” pattern—Jamaat draws support from both the poorest and wealthiest constituencies.

TABLE VI: Deductive Hypothesis Testing Summary

H#	Hypothesis	Verdict
1	Elite/Wealth Reversal (U-shaped)	Confirmed
2	Senior Citizen Firewall	Rejected
3	Student/Youth Vanguard	Partial
4	Referendum Generational Chasm	Rejected
5	Gender Partisan Divide	Partial
6	Digital Battleground	Rejected
7	Border Radicalization	Confirmed
8	Coalition Referendum Cleavage	Confirmed
9	Digital Referendum Paradox	Weak
10	Poverty Anti-Reform Backlash	Weak

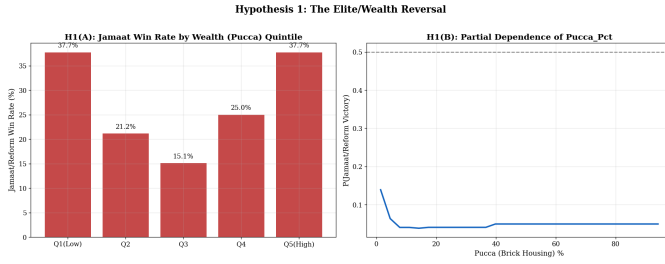


Fig. 8: H1: U-shaped Jamaat win rate by wealth quintile.

B. H2: Senior Citizen Firewall

BNP mean senior percentage: 13.22%; Jamaat: 13.77%. $t = -1.5284$, $p = 0.1289$ —not significant. **Rejected.**

C. H3–H6: Youth, Generational, Gender, Digital

Student Vanguard Index importance = 0.0659 (H3, partial). Referendum generational correlations negligible: Senior vs. NO $r = 0.0062$, Youth vs. YES $r = -0.0341$ (H4, rejected). Female_Pop_Pct importance = 0.0469 (H5, partial). Internet_Pct importance controlling for turnout = 0.0117 (H6, rejected).

D. H7: Border Radicalization

Border Jamaat win rate: **34.1%** vs. interior: 21.4%. Odds ratio = **1.90**. Border constituencies are nearly twice as likely to elect Jamaat candidates. **Confirmed.**

E. H8: Coalition Referendum Cleavage

BNP mean NO_Pct: **34.62%**; Jamaat: 29.44%. $p = 0.000781$ —highly significant at $\alpha = 0.001$. **Confirmed.**

F. H9–H10: Digital Referendum and Poverty Backlash

Internet vs. YES: $r = 0.0952$ (H9, weak). Poverty vs. NO: $r = 0.0816$ (H10, weak). Both directions are consistent with hypotheses but magnitudes are insufficient for confirmation.

G. Advanced Controversial Hypotheses

Hypothesis A (Horseshoe Wealth): Decile-level analysis confirms the U-shaped Jamaat pattern at higher resolution.

Hypothesis B (Digital Elitism): Regression coefficient for unconnected youth predicting NO vote: $\beta = +0.3479$. Digitally excluded youth lean toward referendum rejection.

Hypothesis C (Geopolitical Securitization): Logistic coefficients: Extreme Poverty = -0.0052 (negligible), Border

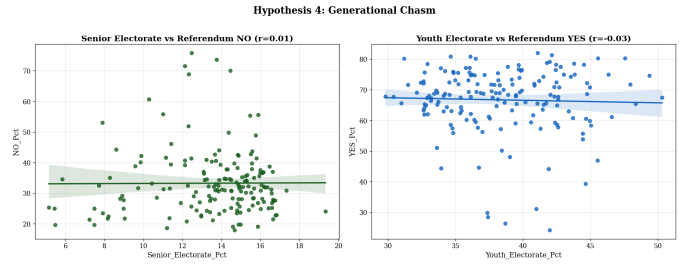
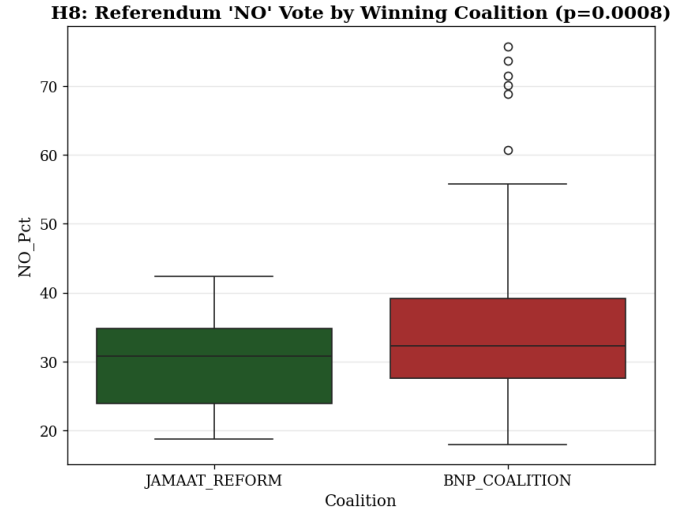


Fig. 9: H4: Referendum outcomes by age-structure. No generational chasm detected.

Fig. 10: H8: Referendum NO vote by coalition ($p = 0.000781$).

Proximity = $+0.6752$. Border effects on Jamaat support are *independent* of poverty.

XI. RESULTS AND DISCUSSION

The pipeline yields six principal findings:

- 1) **Youth as Primary Determinant:** Youth_Electorate_Pct (0.1259) and First_Time_Voter_Pct (0.1201) together account for 24.61% of classifier discriminative power.
- 2) **Internet–Turnout Paradox:** Internet penetration shows the strongest correlation with turnout ($r = -0.5948$) but minimal coalition predictive power (importance = 0.0563)—it affects *whether* people vote, not *whom*.
- 3) **Turnout Predictability vs. Margin Unpredictability:** Turnout $R^2 = 0.6302$; vote share $R^2 = -0.0051$. Participation is structurally determined; mandate strength is campaign-dependent.
- 4) **U-Shaped Wealth Effect:** Both Q1 and Q5 favor Jamaat at 37.74%, while Q3 favors BNP (Jamaat = 15.09%).
- 5) **Border Radicalization:** OR = 1.90, independent of poverty ($\beta_{\text{border}} = +0.6752$ vs. $\beta_{\text{poverty}} = -0.0052$).
- 6) **Coalition–Referendum Cleavage:** The only statistically confirmed partisan divide on the referendum ($p = 0.000781$).

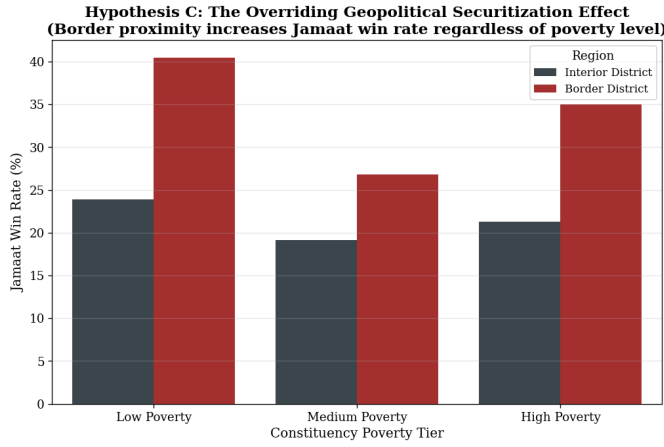


Fig. 11: Hypothesis C: Border proximity ($\beta = +0.6752$) drives Jamaat support independently of poverty ($\beta = -0.0052$).

The data reveals a deeply bifurcated electorate: 87.4% rural-deprived constituencies with high turnout versus 12.6% urban-affluent constituencies with low engagement. Coalition allegiance is primarily determined by generational composition—not wealth, gender, or digital access.

XII. CHALLENGES AND LIMITATIONS

Data Incompleteness: Twelve districts lacked SDG census sheets, requiring KNN imputation. The July Charter referendum data was only confirmed for 168 of 300 constituencies at the time of analysis, limiting the generalizability of referendum-specific findings.

Geospatial Approximation: The BBS administrative hierarchy and Election Commission boundaries do not align perfectly. The “majority proxy” NLP mapping approach achieved 89.6% coverage (269 seats), but excluded 31 constituencies, potentially introducing localized selection bias.

Ecological Inference: All analyses operate at the constituency aggregate level. Correlations do not strictly imply individual-level causal voting behavior.

XIII. CONCLUSION

This paper presents a robust PySpark data pipeline analyzing the 2026 Bangladesh National Election. We successfully integrated heterogeneous BBS census data with EC results across 269 constituencies.

Our findings demonstrate that electoral participation (turnout) is structurally predictable from census data ($R^2 = 0.6302$), primarily driven by the absence of extreme demographic concentrations and an internet-turnout paradox ($r = -0.5948$). Conversely, vote share is unpredictable from demographics ($R^2 = -0.0051$).

The Random Forest coalition classifier establishes youth demographics as the dominant predictor of partisan victory (Acc = 76.79%, AUC = 0.7738). K-Means clustering highlights a severe urban-rural divide. Hypothesis testing confirms a U-shaped Elite/Wealth reversal for Jamaat support, significant

border radicalization (OR = 1.90), and a notable coalition cleavage on the July Charter referendum ($p = 0.000781$).

Future work will expand this pipeline to incorporate 2018 and 2024 historical elections, assessing the temporal stability of these socio-economic determinants, and explore GIS-based polygon intersection to improve geospatial boundary mapping to 100% coverage.

REFERENCES

- [1] Bangladesh Bureau of Statistics, “Population and Housing Census 2022: National Report,” Ministry of Planning, Government of the People’s Republic of Bangladesh, 2023.
- [2] Bangladesh Election Commission, “Results of the 13th National Parliamentary Election, February 12, 2026,” Official Results Portal, 2026. [Online]. Available: <https://www.ecs.gov.bd>
- [3] M. Zaharia *et al.*, “Apache Spark: A Unified Engine for Big Data Processing,” *Commun. ACM*, vol. 59, no. 11, pp. 56–65, 2016.
- [4] X. Meng *et al.*, “MLlib: Machine Learning in Apache Spark,” *J. Mach. Learn. Res.*, vol. 17, no. 34, pp. 1–7, 2016.
- [5] O. Troyanskaya *et al.*, “Missing value estimation methods for DNA microarrays,” *Bioinformatics*, vol. 17, no. 6, pp. 520–525, 2001.
- [6] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.